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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

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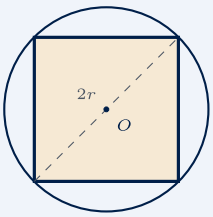
Q1. A square is inscribed in a circle of radius r. The area of the square is: [1]

(A) r^2 (B) $2r^2$ (C) $4r^2$ (D) $\sqrt{2}r^2$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Diagonal of Inscribed Square [0.5 Mark]

The diagonal of a square inscribed in a circle equals the diameter of the circle:



$$\text{Diagonal } d = 2r$$

Step 2: Area of Square from Diagonal [0.5 Mark]

$$\text{Area of square} = \frac{1}{2}d^2 = \frac{1}{2}(2r)^2 = \frac{1}{2}(4r^2) = 2r^2$$

Correct Option: (B) $2r^2$

Area of inscribed square = $\mathbf{2r^2}$

Q2. The area of the incircle of an equilateral triangle of side 42 cm is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 231 cm^2 (B) 462 cm^2 (C) 924 cm^2 (D) 154 cm^2

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Calculate Inradius r [0.5 Mark]

For an equilateral triangle of side $a = 42$ cm:

$$r = \frac{a}{2\sqrt{3}} = \frac{42}{2\sqrt{3}} = \frac{21}{\sqrt{3}} = 7\sqrt{3} \text{ cm}$$

Step 2: Area of Incircle [0.5 Mark]

$$\text{Area} = \pi r^2 = \frac{22}{7} \times (7\sqrt{3})^2 = \frac{22}{7} \times 49 \times 3 = 22 \times 7 \times 3 = 462 \text{ cm}^2$$

Correct Option: (B) 462 cm^2

Area of incircle = **462** cm^2

Q3. The perimeter of a sector of a circle of radius 7 cm with central angle 90° is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 11 cm (B) 22 cm (C) 25 cm (D) 35 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Arc Length of the Sector [0.5 Mark]

$$l = \frac{90^\circ}{360^\circ} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 7 = 11 \text{ cm}$$

Step 2: Total Perimeter of Sector [0.5 Mark]

$$\text{Perimeter} = l + 2r = 11 + 2(7) = 11 + 14 = 25 \text{ cm}$$

Correct Option: (C) 25 cm

Perimeter of sector = **25** cm

Q4. The area of a sector whose perimeter is four times its radius r units is: [1]

(A) $\frac{r^2}{4}$ sq. units (B) r^2 sq. units (C) $2r^2$ sq. units (D) $4r^2$ sq. units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Finding Arc Length l [0.5 Mark]

Given: Perimeter $= l + 2r = 4r \implies l = 4r - 2r = 2r$ units.

Step 2: Sector Area [0.5 Mark]

$$\text{Area of sector} = \frac{1}{2}lr = \frac{1}{2}(2r)(r) = r^2 \text{ sq. units}$$

Correct Option: (B) r^2 sq. units

Area of sector = $\mathbf{r^2}$ sq. units

Q5. **Assertion (A):** If the length of the minute hand of a clock is 7 cm, then the area swept by it in 5 minutes is $\frac{77}{6}$ cm^2.

Reason (R): The length of an arc of a sector of angle θ and radius r is given by $l = \frac{\theta}{360^\circ} \times 2\pi r$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verifying Assertion (A) [0.5 Mark]

In 5 minutes, $\theta = \frac{360^\circ}{60} \times 5 = 30^\circ$.

$$\text{Area} = \frac{30^\circ}{360^\circ} \times \pi r^2 = \frac{1}{12} \times \frac{22}{7} \times 7 \times 7 = \frac{154}{12} = \frac{77}{6} \text{ cm}^2$$

Thus Assertion (A) is **True**.

Step 2: Verifying Reason (R) [0.5 Mark]

The statement $l = \frac{\theta}{360^\circ} \times 2\pi r$ is true, but it provides the formula for arc length, not sector area.

Hence Reason (R) is **True**, but does **not** explain (A).

Correct Option: **(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).**

Correct Option: (**B**)

SECTION B: Very Short Answer Type

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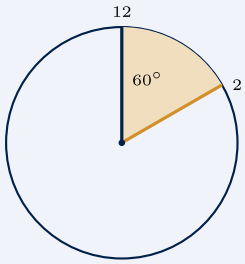
Q6. The minute hand of a clock is 10.5 cm long. Find the area described by the minute hand on the face of the clock between 7:00 am and 7:10 am. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Angle Rotated in 10 Minutes [1 Mark]

Time interval = 7 : 10 − 7 : 00 = 10 minutes.

$$\theta = \frac{360^\circ}{60} \times 10 = 60^\circ$$



Radius $r = 10.5 \text{ cm} = \frac{21}{2} \text{ cm}$.

Step 2: Area Swept by Minute Hand [1 Mark]

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2}$$

$$= \frac{1}{6} \times \frac{22}{7} \times \frac{441}{4} = \frac{1}{6} \times \frac{1386}{4} = \frac{231}{4} = 57.75 \text{ cm}^2$$

Area described = **57.75 cm²** $\left(\text{or } \mathbf{\frac{231}{4}} \text{ cm}^2\right)$

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. An athletic running track has two straight parallel segments of length 106 m each, and the distance between the two inner parallel lines is 60 m. The track is 10 m wide everywhere. Find:

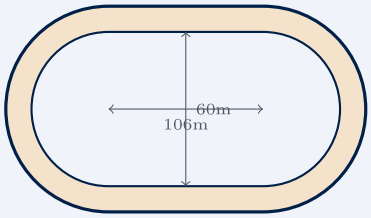
(i) the distance around the track along its inner edge, and

(ii) the total area of the track. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Straight segment length $L = 106$ m. Distance between inner parallel lines $= 60$ m $\implies$ inner radius $r = 30$ m. Track width $w = 10$ m $\implies$ outer radius $R = 30 + 10 = 40$ m.



Part (i): Distance Around Inner Edge [1.5 Marks]

$$\text{Distance} = 2L + 2(\pi r) = 2(106) + 2 \times \frac{22}{7} \times 30$$

$$= 212 + \frac{1320}{7} = \frac{1484 + 1320}{7} = \frac{2804}{7} \text{ m} \approx 400.57 \text{ m}$$

Part (ii): Total Area of the Track [1.5 Marks]

$$\text{Area of 2 rectangular strips} = 2 \times (L \times w) = 2 \times (106 \times 10) = 2120 \text{ m}^2$$

$$\text{Area of 2 semicircular ends} = \pi(R^2 - r^2) = \frac{22}{7} \times (40^2 - 30^2)$$

$$= \frac{22}{7} \times (1600 - 900) = \frac{22}{7} \times 700 = 2200 \text{ m}^2$$

$$\text{Total Area of Track} = 2120 + 2200 = 4320 \text{ m}^2$$

$$\boxed{\text{(i) Inner Distance} = \frac{\mathbf{2804}}{\mathbf{7}} \text{ m} \approx \mathbf{400.57} \text{ m}, \quad \text{(ii) Total Area} = \mathbf{4320} \text{ m}^{\mathbf{2}}}$$

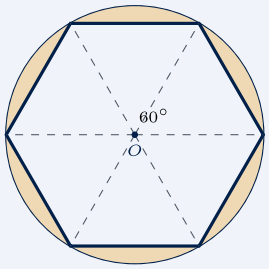
Q8. A round table cover has six equal designs. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of Rs. 0.35 per cm^2. $\left(\text{Use } \sqrt{3} = 1.7, \pi = \frac{22}{7}\right)$ [3]

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► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Radius $r = 28$ cm, Number of designs $n = 6$.

$$\text{Central angle } \theta = \frac{360^\circ}{6} = 60^\circ$$



Step 1: Area of One Sector and One Triangle [1 Mark]

$$\text{Area of 1 sector} = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 28 \times 28 = \frac{1232}{3} \text{ cm}^2$$

$$\text{Area of equilateral } \triangle = \frac{\sqrt{3}}{4} \times (28)^2 = 196\sqrt{3} = 196 \times 1.7 = 333.2 \text{ cm}^2$$

Step 2: Total Area of 6 Designs [1 Mark]

$$\text{Total Area} = 6 \times \left(\frac{1232}{3} - 333.2\right) = 2464 - (6 \times 333.2)$$

$$= 2464 - 1999.2 = 464.8 \text{ cm}^2$$

Step 3: Total Cost Calculation [1 Mark]

$$\text{Cost} = 464.8 \times \text{Rs. } 0.35 = \text{Rs. } 162.68$$

Total Area of Designs = **464.8 cm²**, Total Cost = Rs. **162.68**

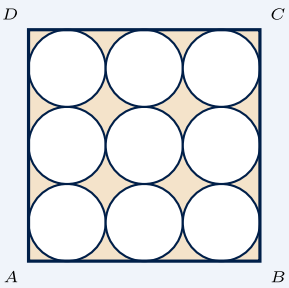
Q9. On a square handkerchief of side 42 cm, nine circular designs each of radius 7 cm are printed in three rows of three circles each. Find the area of the remaining portion of the handkerchief. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Total Area of Square Handkerchief [1 Mark]

Side of square $a = 42$ cm.

$$\text{Area of square} = a^2 = 42 \times 42 = 1764 \text{ cm}^2$$



Step 2: Total Area of 9 Circular Designs [1 Mark]

Radius of each circle $r = 7$ cm.

$$\text{Area of 1 circle} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

$$\text{Area of 9 circles} = 9 \times 154 = 1386 \text{ cm}^2$$

Step 3: Area of Remaining Portion [1 Mark]

$$\text{Remaining area} = \text{Area of square} - \text{Area of 9 designs}$$

$$= 1764 - 1386 = 378 \text{ cm}^2$$

Area of remaining portion = **378 cm²**

SECTION D: Case-Based / Applied Situation MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Case Study: Jeweller's Filigree Silver Brooch Architecture

A goldsmith crafts a silver filigree brooch in the form of a circle of diameter $d = 35$ mm. Thin silver wire is used for the circular rim and also to forge five diameters that divide the brooch into ten equal sectors.

(i) Find the central angle subtended by each individual sector of the brooch. **[1]**

(ii) Calculate the total length of the silver wire required to manufacture the brooch. **[1]**

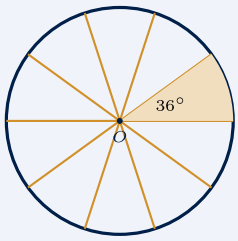
(iii) Calculate the surface area of each sector of the brooch. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

— OR —

Calculate the total circular face area of the brooch. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Diameter $d = 35$ mm $\implies r = \frac{35}{2}$ mm. Number of sectors $= 10$.



Part (i): Central Angle of Each Sector [1 Mark]

$$\theta = \frac{360^\circ}{10} = 36^\circ$$

Part (ii): Total Wire Length Required [1 Mark]

$$\text{Length} = \text{Circumference} + 5 \times d = \pi d + 5d$$
$$= \left(\frac{22}{7} \times 35\right) + (5 \times 35) = 110 + 175 = 285 \text{ mm}$$

Part (iii): Area of Each Sector [2 Marks]

$$\text{Area} = \frac{1}{10} \times \pi r^2 = \frac{1}{10} \times \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = \frac{385}{4} = 96.25 \text{ mm}^2$$

— OR (Alternative Choice) —

Total Circular Face Area of Brooch: [2 Marks]

$$\text{Area} = \pi r^2 = \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = \frac{1925}{2} = 962.5 \text{ mm}^2$$

(i) $\theta = \mathbf{36}^\circ$, (ii) Wire Length $= \mathbf{285}$ mm, (iii) Sector Area $= \mathbf{96.25}$ mm^2 $\left(\text{OR Face Area} = \mathbf{962.5} \text{ mm}^2\right)$

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